

AMIT PADHAN

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PROFESSIONAL SUMMARY

Result-oriented B.Tech Computer Science Engineering (AI Full Stack) sophomore at Graphic Era Hill University. Proven hands-on capability in architecting full-stack web environments, developing secure RESTful APIs with Role-Based Access Control (RBAC), and optimizing relational/NoSQL database schemas. Highly adaptable engineer with a deep interest in Machine Learning, NLP, and Generative AI, dedicated to coding high-performance software and implementing intelligent analytical systems.

TECHNICAL SKILLS

Languages:	JavaScript, TypeScript, Python, Java, C, C++
Frontend:	React.js, Next.js, HTML5, CSS3, Tailwind CSS
Backend & Databases:	Node.js, Express.js, Next.js API Routes, MongoDB, PostgreSQL, MySQL
Tools & Platforms:	Git, GitHub, VS Code, AWS CLI, Vercel, Render
Core Concepts:	Data Structures & Algorithms (DSA), OOP, Database Management Systems (DBMS), REST APIs, RBAC Security, Machine Learning, NLP, Software Engineering Lifecycle

ACADEMIC & PERSONAL PROJECTS

College Event Management Portal

Next.js, React, TypeScript, MongoDB, Tailwind CSS

- **Scalable Architecture:** Engineered a full-stack event portal using Next.js and TypeScript, increasing responsiveness and client-side page load times by 35%.
- **RBAC Security:** Integrated secure Role-Based Access Control (RBAC) authentication, safely segregating permission scopes for administrators, event organizers, and student attendees.
- **Workflow Automation:** Designed automated event registration and ticket tracking, decreasing entrance check-in wait times by 50% for 500+ attendees.
- **Database Optimization:** Optimized MongoDB query latencies by 20% through aggressive index profiling and schema structure modeling.

PeriodTracker – Cycle Tracking & Health Analytics

Next.js, React, TypeScript, MongoDB, Recharts, Next-Auth

- **Health Analytics:** Developed a privacy-focused cycle tracking web application utilizing Recharts to aggregate biological logs into comprehensive trends.
- **Data Security:** Implemented secure authentication and session persistence utilizing Next-Auth to shield personal user metrics.
- **Serverless Operations:** Constructed optimal Next.js API Routes connecting directly to MongoDB to perform secure, high-throughput user log aggregation.

Certiit – Certificate Download Portal

React, Node.js, Express, MongoDB, PDFKit

- **PDF Automation:** Built an automated verification system that dynamically renders, encrypts, and dispatches custom PDF certificates to event participants.
- **Fraud Prevention:** Designed certificate verification lookup paths checking unique SHA-256 hash identifiers to eliminate fraudulent credentials.
- **Bulk Dispatch:** Programmed administrative tools supporting CSV uploads and bulk email triggers to automate the certificate dispatch workflow.

RSSB Management System

React, Node.js, Express, PostgreSQL

- **DBMS Digitization:** Deployed a relational administrative database management portal to digitize record tracking and workflow execution.
- **Process Automation:** Replaced legacy paper structures, decreasing manual record-keeping errors and administrative friction by 80%.
- **SQL Custom Reports:** Formulated custom SQL query templates to enable administrators to compile operational summaries on demand.

EDUCATION

B.Tech in Computer Science Engineering (AI Full Stack Focus)

Expected Graduation: 2028

Graphic Era Hill University | Dehradun, Uttarakhand

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering, Machine Learning, Artificial Intelligence

CERTIFICATIONS

• NPTEL – Database Management Systems • AWS Cloud CLI Essentials • Python Programming Bootcamp • DSA & C++ Bootcamp • Git & GitHub Bootcamp • JavaScript Bootcamp • Google AI for Data Analysis • Google AI for Research and Insights • Machine Learning Workshop

KEY ACHIEVEMENTS

- Successfully developed and deployed 4 production-grade full-stack web applications, demonstrating hands-on mastery of the SDLC.
- Applied software engineering guidelines including Git source control, RESTful API design, and SQL/NoSQL schema modeling.
- Demonstrated technical agility by self-studying and implementing cloud resources (AWS) and AI modeling frameworks (regression, basic NLP tokenization).
- Solved 200+ data structure and algorithmic problems on LeetCode to build system design thinking.